Hybrid AI-based Detection of fileless malware in IOMT Healthcare systems

R26-CS-006

Project Proposal Report

Chathuranga L.A.S – IT22108272

B.Sc. (Hons) Degree in Information Technology Specialized in Cybersecurity

Department of Computer systems and Engineering

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Shakya W.A. R	IT22210142	
Fernando M.S. V	IT22080530	
Pagodaarachchi R. T.	IT22135070	
Chathuranga L.A. S	IT22108272	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Mr. Kanishka Yapa.	13/03/2026	
Mr. Dinith Primal.	13/03/2026	

ABSTRACT (Maximum 300 words)

The rapid expansion of Internet Medical Things (IoMT) technologies has significantly improved the quality and efficiency of healthcare delivery by enabling real-time monitoring, intelligent diagnostics, and integrated medical data management. Among these technologies, medical imaging infrastructures such as Picture Archiving and Communication Systems (PACS) play a critical role in storing, retrieving, and transmitting diagnostic images using the Digital Imaging and Communications in Medicine (DICOM) protocol. However, the increasing connectivity of imaging devices, radiology workstations, and hospital networks has also enlarged the attack surface of healthcare environments, making them more vulnerable to cyberattacks. Healthcare continues to be the costliest industry for data breaches, with IBM reporting an average breach cost of USD 9.77 million in 2024.

Although traditional security mechanisms such as firewalls and rule-based intrusion detection systems are widely deployed in hospitals, they are often ineffective in identifying abnormal or previously unseen communication behaviors in DICOM traffic. Existing research has highlighted PACS and DICOM security weaknesses, including exposure to denial-of-service conditions, insecure configurations, and protocol-specific risks, yet practical anomaly detection mechanisms tailored for DICOM communication remain limited.

To address this gap, this research proposes a machine learning-based anomaly detection component focused on DICOM medical imaging network traffic. The component captures DICOM communication packets, extracts behavioral traffic features, and applies anomaly detection techniques to identify suspicious operations such as unusual C-STORE, C-FIND, and C-MOVE behaviors. The solution will be developed using Python, packet capture tools, and machine learning libraries such as Scikit-learn. The expected outcome is a lightweight and intelligent monitoring mechanism capable of generating timely alerts for anomalous imaging traffic, thereby strengthening the cybersecurity posture of healthcare imaging environments.

Keywords: IoMT, DICOM, PACS, anomaly detection, machine learning, healthcare cybersecurity.

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I would also like to thank the academic staff of the Department of Computer Systems and Engineering at Sri Lanka Institute of Information Technology for providing the knowledge, resources, and motivation required for this research. In addition, I would like to acknowledge my team members for their support, discussions, and collaborative ideas during the early stages of the project. Their insights have been highly valuable in understanding the broader scope of the overall research and positioning this component within the full system.

TABLE OF CONTENT

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 6

LIST OF FIGURES 7

LIST OF ABBREVIATIONS 8

LIST OF APPENDICES 9

1. INTRODUCTION 10

2. OBJECTIVE 16

3. METHODOLOGY 17

4. HIGH-LEVEL SYSTEM ARCHITECTURE 19

5. USER REQUIREMENTS 24

6. COMMERCIALIZATION PLAN 26

7. BUDGET AND JUSTIFICATION 28

8. WORK BREAKDOWN STRUCTURE (WBS) 29

9. GANTT CHART 31

10. REFERENCES LIST 33

11. APPENDICES 35

LIST OF TABLES

- Table 1: Summary of Existing Research Related to DICOM and Healthcare Network Security
- Table 2: Comparison Between Existing Approaches and Proposed Component
- Table 3: Evaluation Metrics
- Table 4: Work Breakdown Structure for Component 1

LIST OF FIGURES

- Figure 1: Proposed workflow of Component 1 – DICOM Medical Imaging Network Anomaly Detection
- Figure 2: High-Level System Architecture of Component 1
- Figure 3: Gantt Chart for Component 1

LIST OF ABBREVIATIONS

AI – Artificial Intelligence

API – Application Programming Interface

CT – Computed Tomography

DICOM – Digital Imaging and Communications in Medicine

F1 – Harmonic Mean of Precision and Recall

IDS – Intrusion Detection System

IoMT – Internet of Medical Things

IP – Internet Protocol

LAN – Local Area Network

ML – Machine Learning

MRI – Magnetic Resonance Imaging

PACS – Picture Archiving and Communication System

TCP – Transmission Control Protocol

XAI – Explainable Artificial Intelligence

LIST OF APPENDICES

- Appendix A – Component 1 Gantt Chart
- Appendix B – Proposed High-Level Architecture Diagram
- Appendix C – Dataset Description
- Appendix D – Workflow Diagram
- Appendix E – Risk Analysis

1. INTRODUCTION

1.1 Background and Context

The healthcare sector has experienced a significant digital transformation over the past decade through the increasing integration of Internet of Medical Things (IoMT) devices, smart monitoring equipment, electronic health platforms, and digital imaging systems. One of the most important technologies in this transformation is the Picture Archiving and Communication System (PACS), which is used to store, manage, retrieve, and transmit medical images generated from diagnostic equipment such as CT scanners, MRI machines, ultrasound devices, and radiology workstations.

PACS communication is mainly carried out using the Digital Imaging and Communications in Medicine (DICOM) protocol. DICOM is the core communication standard that allows imaging devices and hospital systems to exchange diagnostic images and associated metadata. Because DICOM-based medical imaging infrastructure is deeply integrated into hospital operations, any compromise in this communication environment may affect diagnosis, treatment planning, system availability, and patient safety. Research on PACS cybersecurity shows that medical imaging systems remain exposed to risks such as insecure network exposure, malware propagation, denial-of-service conditions, and malicious manipulation of imaging workflows.

At the same time, healthcare organizations remain one of the most attractive targets for cybercriminals because they manage sensitive personal data, essential services, and time-critical operations. IBM's 2024 Cost of a Data Breach report states that healthcare had the highest average breach among all industries, at USD 9.77 million.

Despite these risks, many hospital environments still rely primarily on conventional security tools such as firewalls, signature-based intrusion detection systems, and static rule-based monitoring. These mechanisms can detect known threats, but they often fail to identify unusual communication patterns that do not match predefined signatures. This is particularly problematic in DICOM environments, where traffic often appears legitimate at the protocol level even when behavior is abnormal.

This component therefore proposes a machine learning-based anomaly detection mechanism for DICOM medical imaging traffic. The purpose is to capture DICOM packets, model normal behavior patterns, and identify deviations that may indicate suspicious activities such as unusual C-STORE uploads, irregular C-MOVE requests, excessive querying, unauthorized communication attempts, or exfiltration-like behavior in imaging networks.

This research aligns with **Sustainable Development Goal 3: Good Health and Well-Being**, as it contributes to safer and more reliable healthcare infrastructure. It also aligns with **Sustainable Development Goal 9: Industry, Innovation and Infrastructure**, as it supports secure digital innovation in critical systems.

This work further aligns with the **Information Assurance and Security** research cluster by focusing on intelligent protection mechanisms for healthcare communication environments and improving the resilience of critical medical systems against cyber threats.

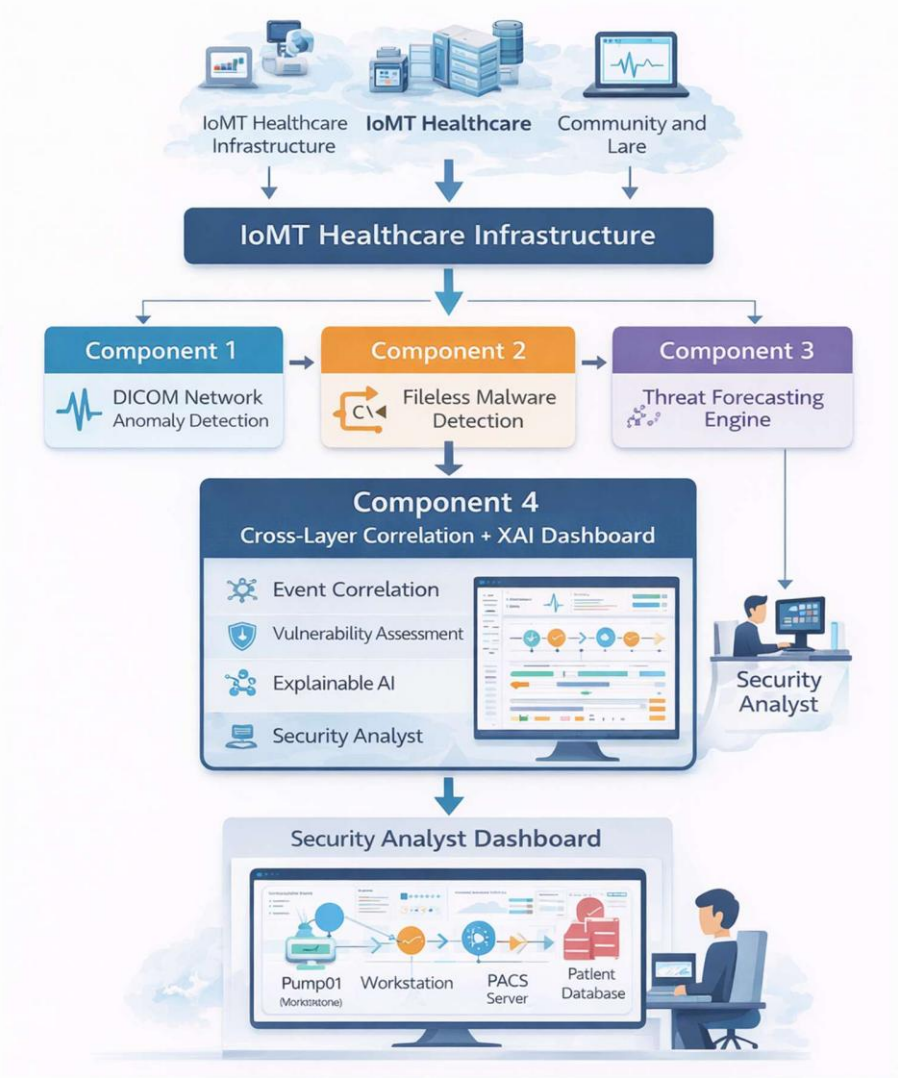


Figure 1: Hybrid AI-driven multi-layer security framework workflow diagram.

1.2 Literature Review

Recent literature demonstrates increasing concern regarding the cybersecurity posture of healthcare systems, particularly those involving Internet of Medical Things (IoMT) devices, Picture Archiving and Communication Systems (PACS), and medical imaging networks. As healthcare infrastructures become more interconnected and digitally dependent, the attack surface available to malicious actors continues to grow. Unlike traditional IT systems, healthcare environments combine general-purpose computing systems, specialized medical devices, legacy equipment, and time-sensitive clinical workflows. This makes them highly complex and difficult to secure. In this context, many researchers have emphasized that healthcare cybersecurity can no longer be approached using conventional enterprise-only security models, because medical environments introduce protocol-specific, device-specific, and safety-critical challenges that demand specialized attention.

Adil et al. identify healthcare IoT ecosystems as highly heterogeneous and vulnerable to a broad range of cyber threats, including unauthorized access, malware infections, ransomware, data interception, denial-of-service attacks, and advanced persistent attacks that specifically target interconnected medical infrastructure. Their work is valuable because it provides a broad and up-to-date understanding of the overall threat landscape in IoMT-based healthcare systems. It explains how the growing dependence on connected healthcare technologies improves operational efficiency and patient care, while simultaneously creating serious security risks. The study also highlights the fact that many IoMT devices are resource-constrained, difficult to patch, and often deployed with weak security controls, making them attractive targets for attackers. Although this research provides an important general security background for healthcare IoT, it does not focus specifically on the behavioral analysis of DICOM traffic or the detection of anomalies in PACS communications. Therefore, while it establishes the seriousness of the problem, it does not directly address the protocol-level detection requirements of this component.

Eichelberg's overview of PACS and medical imaging cybersecurity further strengthens the argument that imaging networks require dedicated security attention. The study explains that PACS-specific security cannot be viewed in isolation and must instead be considered together with the broader hospital IT environment, including servers, databases, user workstations, network devices, and medical imaging modalities. The paper highlights several practical and operational risks, such as insecure deployment configurations, exposed imaging services, malicious image manipulation, data leakage, and protocol misuse. This literature is especially important because it confirms that medical imaging systems are not only theoretically vulnerable but also operationally exposed to real cybersecurity risks. In addition, the paper makes it clear that attacks on medical imaging systems can have consequences beyond data loss, including diagnostic disruption, delays in treatment, and direct patient safety implications. However, despite presenting a useful overview of the threat landscape, the study remains descriptive in nature and does not propose a dedicated anomaly detection framework for monitoring DICOM traffic behavior in real time. As a result, it identifies the problem domain clearly but leaves a technical implementation gap that this research aims to address.

Wang et al. explored DICOM vulnerability mining using fuzzing techniques and demonstrated that open-source DICOM libraries may contain weaknesses that can lead to protocol misuse, denial-of-service conditions, parsing failures, and other exploitable behaviors. This study is significant because it offers technical evidence that DICOM communication stacks themselves can become a point of attack, even when the surrounding infrastructure appears legitimate. It shows that DICOM is not merely a passive transmission format, but rather a protocol ecosystem whose implementations may contain serious weaknesses. The findings are especially relevant to this research because they justify the need for monitoring DICOM-related behavior more carefully. If DICOM libraries and services can be exploited or manipulated, then abnormal protocol interactions may become an important indicator of malicious activity. Nevertheless, the main contribution of Wang et al. is vulnerability discovery rather

than runtime anomaly detection. Their work focuses on identifying weaknesses in implementations under test conditions, rather than modeling normal versus abnormal DICOM behavior in a live or simulated network environment. Therefore, while highly relevant from a vulnerability perspective, it does not provide a direct detection mechanism suitable for operational PACS monitoring.

Hady et al. compared healthcare intrusion detection systems built using medical data and network data and concluded that combining multiple data perspectives can improve detection performance. Their work clarifies the broader design considerations of IDS solutions within healthcare settings, particularly the trade-offs between data sources, system performance, and detection effectiveness. The study suggests that healthcare security monitoring becomes stronger when detection models do not rely on a single type of input alone. This idea is useful for the full research project because the overall hybrid framework also uses multiple layers. However, from the perspective of Component 1, the limitation is that the study remains broad and does not focus on protocol-specific modeling of DICOM operations such as C-STORE, C-FIND, and C-MOVE. In a medical imaging context, these commands have workflow-specific meanings, and abnormal usage of them may reveal exfiltration attempts, unauthorized queries, misuse of imaging services, or suspicious retrieval behavior. Since Hady et al. discuss healthcare IDS at a higher level, their work supports the general need for improved healthcare intrusion detection, but it does not directly address the specialized traffic characteristics of imaging networks.

Summary of Existing IoMT Security Research

Study	Technique	Focus	Focus	Limitation
Adil et al. (2024)	IoMT security survey	General healthcare IoT	General healthcare IT threats	Not DICOM specific
Eichelberg (2020)	PACS security overview	PACS and imaging	PACS and imaging cybersecurity	No anomaly detection model
Wang et al. (2019)	Fuzzing / vulnerability mining	DICOM protocol	DICOM protocol weaknesses	Not real-time traffic monitoring
Hady et al. (2020)	IDS comparison	IDS comparison	Healthcare IDS using medical and network data.	No DICOM behavioral focus
Karagiannis et al. (2023)	Simulation-based modeling	DICOM/PACS environment	DICOM/PACS test environment	Limited detection analytics
Capuano et al. (2022)	XAI survey	XAI survey	Explainability in cybersecurity	Limited detection analytics
Capuano et al. (2022)	XAI survey	XAI survey	Explainability in cybersecurity	Not imaging-network specific

Table 1: Summary of Existing IoMT Security Research

Karagiannis et al. presented a simulation-based approach for DICOM communication and PACS integration, offering a useful environment for cybersecurity experimentation involving medical imaging workflows. This contribution is particularly important because obtaining real-world DICOM datasets from hospitals is difficult due to privacy, confidentiality, legal, and operational constraints. As a result, researchers often struggle to obtain realistic traffic data for experimentation. The simulation-based method proposed by Karagiannis et al. is therefore highly relevant, since it offers a practical strategy for generating controlled DICOM communication scenarios that resemble real healthcare workflows. Such environments can be used to model image transfers, storage operations, retrieval requests, and system interactions in a reproducible manner. For the present research, this is valuable because it provides a foundation for creating a dataset suitable for anomaly detection experiments. However, the study is primarily centered on simulation and practical demonstration rather than anomaly detection modeling. It helps solve the data generation problem, but not the problem of how to identify suspicious behavioral patterns in the resulting communication flows.

Recent work on machine learning-based anomaly detection in broader security contexts suggests that unsupervised and semi-supervised learning models are particularly suitable for environments where attack signatures are incomplete, constantly evolving, or entirely unknown. In cybersecurity domains

where new threats emerge rapidly, it is often unrealistic to assume that all malicious behaviors can be captured in labeled datasets. This is especially true in DICOM traffic analysis, where abnormal behavior may not always correspond to a known exploit, malware family, or fixed rule violation. Instead, suspiciousness may emerge from subtle deviations in timing, flow frequency, packet size patterns, connection sequences, protocol command usage, or unexpected device-to-device communication. In such cases, anomaly detection methods become highly valuable because they focus on learning what "normal" behavior looks like and then identifying deviations from that baseline. This is an important conceptual foundation for the proposed research, as hospitals may not always know in advance how abnormal DICOM activity will appear. Consequently, a behavior-based anomaly detection approach is more practical than a purely signature-based method for this specific context.

Overall, the literature shows four major themes: general IoMT security, PACS and DICOM vulnerabilities, healthcare IDS design, and explainable AI for cybersecurity. Each of these research directions contributes something important to the understanding of the problem. General IoMT security literature explains the scale and seriousness of threats in connected healthcare. PACS and DICOM studies confirm that imaging systems are realistic and valuable targets. Healthcare IDS literature demonstrates the need for intelligent monitoring beyond traditional rules. XAI literature highlights the importance of trust and interpretability in operational security systems. However, despite these contributions, there remains limited work focused specifically on real-time or near real-time anomaly detection for DICOM communication behavior in PACS environments. This gap justifies the need for the proposed component, which aims to develop a DICOM-aware anomaly detection mechanism tailored to healthcare imaging networks.

1.3 Research Gap

Clear statement of what is missing

Although prior research has examined healthcare cybersecurity, PACS vulnerabilities, DICOM protocol weaknesses, and machine learning-based intrusion detection, there is still a lack of healthcare-specific anomaly detection mechanisms that focus directly on **behavioral monitoring of DICOM communication traffic** within PACS environments.

Why current approaches are insufficient

Current security approaches in healthcare imaging environments are insufficient for three main reasons. First, many traditional systems are rule-based and therefore depend on known signatures or fixed thresholds. Second, many healthcare IDS solutions analyze traffic at a generic network level without understanding the operational context of DICOM commands. Third, vulnerability-focused research identifies protocol and implementation weaknesses but does not necessarily monitor whether live DICOM communication is behaving abnormally in practice.

In a real hospital environment, an attacker or misconfigured system may generate traffic that is syntactically valid DICOM traffic but behaviorally suspicious, such as unusual image transfer rates, repeated query operations, off-hours data movement, abnormal source-destination communication, or excessive retrieval behavior. These activities may bypass traditional tools if they appear legitimate at the packet level.

Justification of novelty and creativity

The novelty of this component lies in building a **DICOM-aware anomaly detection module** that learns normal imaging communication behavior and identifies deviations in real time or near real time.

Rather than treating medical imaging traffic as ordinary network traffic, this component gives protocol-specific attention to PACS-related workflows and operation patterns.

Its creativity comes from combining packet capture, feature engineering, DICOM behavioral analysis, and machine learning-based anomaly detection into one healthcare-specific module that can later integrate with the rest of the hybrid AI framework.

Evidence-based argument

Existing literature confirms that PACS and DICOM infrastructures remain vulnerable, that healthcare systems require better detection mechanisms, and that anomaly detection is useful in dynamic environments where signature-based methods are insufficient. However, the current body of work does not adequately deliver a DICOM-centered anomaly detection component tailored for operational hospital imaging networks.

2. OBJECTIVE

2.1 Main Objective (Measurable)

The main objective of this component is to design and evaluate a machine learning-based anomaly detection mechanism capable of identifying abnormal DICOM medical imaging network traffic within PACS-based healthcare environments. The proposed solution aims to monitor communication patterns between medical imaging devices, radiology workstations, and PACS servers, and to detect suspicious deviations that may indicate unauthorized access, misuse of imaging services, abnormal data transfers, or potential cyber threats. In measurable terms, the component will be assessed based on its ability to detect anomalous DICOM traffic with acceptable performance using evaluation metrics such as accuracy, precision, recall, and F1-score.

2.2 Specific Objectives (4–6 measurable)

Objective1

To collect and analyze DICOM traffic generated by medical imaging devices, radiology workstations, and PACS systems in a controlled healthcare network environment.

Objective 2

To preprocess capture DICOM packet data and extract behavioral features such as packet size, session duration, communication frequency, time intervals, and DICOM operation patterns.

Objective 3

To develop and train a machine learning-based anomaly detection model capable of distinguishing normal DICOM communication behavior from abnormal traffic patterns.

Objective 4

To generate anomaly alerts for suspicious DICOM activities such as unusual C-STORE, C-FIND, and C-MOVE operations.

Objective 5

To evaluate the proposed component using standard performance metrics such as accuracy, precision, recall, F1-score, and false positive rate.

Objective6

To provide outputs that can be integrated with the overall hybrid healthcare security framework for broader cross-layer analysis.

3. METHODOLOGY

Research approach

This study follows a System Development and Experimental Evaluation approach, which is commonly used in information technology and cybersecurity research when developing new security mechanisms or intelligent detection systems. The main purpose of this approach is to design, implement, and experimentally validate a functional anomaly detection prototype capable of monitoring DICOM medical imaging traffic within PACS-based healthcare environments.

The research process begins with the conceptual design of the anomaly detection architecture, followed by the development of a prototype system capable of capturing and analyzing DICOM network traffic. After the system design phase, simulated or controlled network traffic will be generated or collected to create realistic communication scenarios between medical imaging devices and PACS servers. The captured data will then undergo preprocessing and feature extraction processes to transform raw packet information into structured datasets suitable for machine learning analysis.

Subsequently, machine learning algorithms will be applied to identify abnormal communication patterns that deviate from normal DICOM network behavior. The developed model will then be evaluated through controlled experiments using both normal and abnormal traffic scenarios. Finally, the performance of the system will be assessed using predefined evaluation metrics to determine the effectiveness and reliability of the proposed anomaly detection component.

Technical methods and algorithms.

The technical implementation of this component is based on a multi-stage data processing and anomaly detection pipeline consisting of five main stages.

- First, DICOM network traffic will be captured from a simulated or controlled PACS environment using packet capture tools such as Wireshark or tcpdump. These tools will allow the system to monitor network packets exchanged between medical imaging devices, radiology workstations, and PACS servers. Capturing the traffic at this stage ensures that the dataset accurately reflects real communication patterns within medical imaging infrastructures.
- Second, the captured packets will be parsed and analyzed to identify features relevant to DICOM communication behavior. These features may include packet length, communication session duration, time-based intervals between packets, source and destination IP addresses, frequency of association requests, and the occurrence of DICOM protocol operations such as C-STORE, C-FIND, and C-MOVE commands. These behavioral features help represent the characteristics of imaging communication flows.
- Third, the extracted data will undergo preprocessing and cleaning operations. During this stage, irrelevant information will be removed, missing values will be handled, and the data will be normalized to ensure consistency. The processed data will then be transformed into a structured dataset that can be used as input for machine learning model training.
- Fourth, anomaly detection algorithms such as Isolation Forest, One-Class Support Vector Machine (One-Class SVM), or Local Outlier Factor (LOF) will be explored for detecting abnormal traffic patterns. Among these methods, Isolation Forest is expected to be the primary algorithm because it is computationally efficient, capable of handling large datasets, and particularly suitable for unsupervised anomaly detection tasks where labeled attack data may be limited.

Finally, the trained model will analyze incoming feature vectors and assign anomaly scores to traffic

flows. When the anomaly score exceeds a predefined threshold, the system will generate alerts indicating suspicious or abnormal network behavior within the DICOM communication environment.

Tools and platforms (with justification)

The implementation of the proposed prototype will utilize a combination of **cybersecurity tools, machine learning libraries, and network traffic analysis platforms**. The main development language selected for this research is **Python**, due to its strong ecosystem of libraries for data analysis, machine learning, and cybersecurity experimentation.

The primary tools and libraries used in the implementation include:

- **Wireshark / tcpdump** for network packet capture and traffic monitoring
- **Python** for implementing the anomaly detection system and processing pipeline
- **Pandas and NumPy** for data preprocessing, cleaning, and feature engineering
- **Scikit-learn** for implementing machine learning algorithms such as Isolation Forest and One-Class SVM
- **Matplotlib or Plotly** for visualizing experimental results and anomaly detection outputs
- **Jupyter Notebook or Visual Studio Code (VS Code)** for development and experimentation
- **VirtualBox or VMware** to simulate a PACS network environment and generate controlled DICOM traffic scenario.

Data collection and ethical considerations

The proposed system will be validated using a combination of **simulated traffic datasets and controlled experimental scenarios**. Since obtaining real hospital network traffic is extremely difficult due to privacy and security regulations, the research will primarily rely on a laboratory simulation environment.

The data set used in this study will include:

- **Simulated DICOM traffic generated in a controlled PACS environment**
- **Healthcare IoT network datasets** used to simulate background medical network traffic
- **Synthetic abnormal traffic scenarios** designed to represent suspicious behaviors such as abnormal image transfers, unusual request patterns, or irregular communication flows

Ethical considerations are an important aspect of this research. No real patient data, medical records, or personal identifiable information will be collected or used during the experimentation process. All testing will be conducted in an isolated laboratory environment to prevent interference with real healthcare systems. Any packet captures generated during experimentation will be anonymized if necessary to ensure compliance with data protection principles.

Validation strategy and metrics

The performance of the proposed anomaly detection component will be evaluated through **controlled experimental testing** involving both normal and abnormal DICOM communication scenarios. These experiments will allow the system to be tested under different traffic conditions to measure its ability to accurately identify suspicious network behavior.

To evaluate the effectiveness of the detection model, several standard machine learning performance metrics will be used.

The evaluation metrics include:

- **Accuracy** – measures the overall correctness of the anomaly detection model
- **Precision** – measures how many detected anomalies are actually malicious or abnormal
- **Recall** – measures the model's ability to detect true anomalies in the dataset
- **F1 Score** – provides a balanced measure between precision and recall
- **False Positive Rate** – measures how often the system incorrectly flags normal traffic as anomalous
- **Detection Latency** – measures how quickly the system can detect abnormal traffic patterns after they occur

These metrics will help determine whether the proposed anomaly detection component is capable

of accurately identifying abnormal DICOM communication patterns while minimizing false alerts. The results obtained from these evaluations will be compared with baseline detection approaches to assess the overall effectiveness of the proposed system.

Metric	**Purpose**
Accuracy	Overall detection performance
Precision	Ratio of correct anomaly alerts
Recall	Ability to detect true abnormal traffic
F1 Score	Balance between precision and recall
False Positive Rate	Incorrect anomaly alerts
Detection Latency	Speed of anomaly identification

Table 3: Evaluation metrics.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed solution is a **DICOM medical imaging network anomaly detection component** designed for PACS-based healthcare environments. The purpose of this component is to monitor communication between medical imaging devices, radiology workstations, and PACS servers to identify abnormal traffic patterns that may indicate potential cybersecurity threats. The system captures DICOM network traffic, extracts behavioral communication features, and applies machine learning-based anomaly detection models to detect suspicious activities such as unusual data transfers, abnormal query requests, or unexpected communication patterns within the imaging network.

The architecture of the proposed component consists of several functional submodules responsible for traffic capture, preprocessing, feature extraction, anomaly detection, and alert generation. These modules work together to analyze DICOM traffic behavior and identify deviations from normal communication patterns. The alerts generated can assist healthcare IT administrators and security analysts in detecting potential threats at an early stage and improving the overall security posture of medical imaging infrastructures.

The major subcomponents are:

1. **Traffic Source Layer**
   Includes CT scanners, MRI machines, radiology workstations, PACS servers, and other DICOM-enabled medical systems.

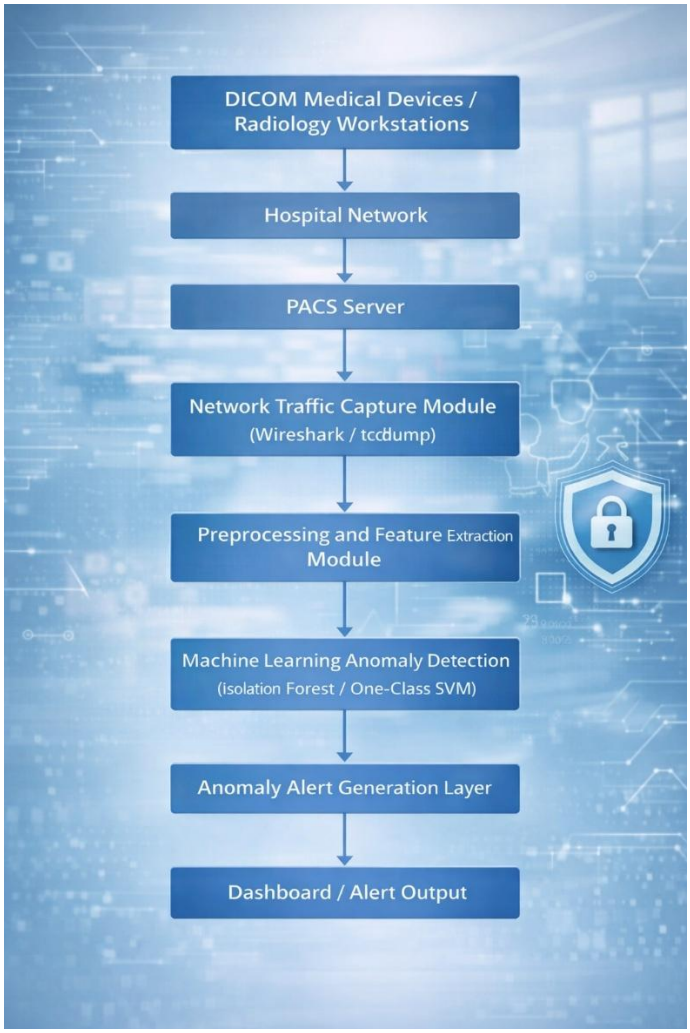
2. **Packet Capture Module**
   Collects network traffic relevant to DICOM communication.

3. **Preprocessing and Feature Extraction Module**
   Transforms packet-level data into structured behavioral features.

4. **Anomaly Detection Engine**
   Uses machine learning algorithms to detect deviations from normal traffic patterns.

5. **Alert and Reporting Layer**
   Produces warnings and summaries for detected anomalies

4.2 System Diagram of Components



4.3 Highlighting the Individual Component and Its Parts

This component contains the following main parts:

a) Input Sources

- DICOM traffic from imaging devices
- PACS communication flows
- radiology workstation interactions
- network session metadata

b) Capture and Parsing Layer

- packet collection
- traffic filtering
- DICOM flow identification
- parsing of protocol-related behaviors

c) Feature Engineering Layer

- packet size distribution
- communication frequency
- session timing
- operation counts
- source-destination behavior

d) Detection Layer

- Normally Profile Learning
- anomaly scoring
- threshold-based alert triggering

e) Output Layer

- anomaly alert logs
- suspicious flow summaries
- visualization-ready outputs

4.4 Component Interactions

Component 4 connects with the previous research components and the healthcare ecosystem.

Component 1 operates as the medical imaging network monitoring layer within the overall hybrid AI-driven healthcare security framework.

It **captures and analyzes raw DICOM network traffic** generated by medical imaging devices such as CT scanners, MRI machines, radiology workstations, and PACS servers.

The component **monitors communication behavior between imaging devices and PACS infrastructure** within the hospital network.

Captured traffic is processed through **packet capture, preprocessing, and feature extraction modules** to prepare the data for anomaly detection.

The **machine learning anomaly detection model analyzes DICOM traffic patterns** to identify abnormal or suspicious communication behavior.

When unusual traffic patterns are detected, the system **generates anomaly alerts related to DICOM operations** such as abnormal C-STORE, C-FIND, or C-MOVE requests.

The alerts generated by this component **serve as input signals for higher-level components** in the overall security framework.

The output of Component 1 can be **correlated with endpoint-level security events from Component 2**, which focuses on PowerShell-based fileless malware detection.

The anomaly results can also be **combined with threat forecasting insights from Component 3** to improve proactive threat prediction.

Finally, the outputs are **integrated with Component 4 for cross-layer correlation and XAI-based visualization**, enabling analysts to understand how imaging network anomalies relate to broader cyberattack patterns.

4.5 Data Flow Explanation

1) Traffic Collection

In this stage, DICOM network traffic is captured from the simulated or monitored healthcare network environment. Packet capture tools such as Wireshark or tcpdump are used to collect communication data between medical imaging devices, radiology workstations, and PACS servers. The captured traffic includes various DICOM operations such as image storage, query, and retrieval requests that occur during normal medical imaging workflows.

2) Data Preprocessing

After capturing the raw packet data, the collected traffic is cleaned and organized for further analysis. This step involves filtering out irrelevant packets, removing duplicate records, and normalizing the data format. The processed traffic is then grouped into communication sessions or traffic windows to represent meaningful network interactions between devices.

3) Feature Extraction

Once the data is preprocessed, relevant behavioral features are extracted from the network traffic. These features may include packet size, communication frequency, session duration, time intervals between packets, and types of DICOM commands used during communication. These extracted features help represent the behavior of network traffic in a structured format that can be analyzed by machine learning models.

4) Anomaly Detection

In this stage, the extracted feature vectors are provided as input to the trained machine learning model. Algorithms such as Isolation Forest or One-Class SVM analyze the patterns within the data and compare them against learned normal behavior. If the model identifies unusual patterns that deviate from normal traffic behavior, those patterns are flagged as potential anomalies.

5) Alert Generation

Finally, when the anomaly detection model assigns anomaly scores above a predefined threshold, the system generates alerts indicating suspicious DICOM communication activity. These alerts can then be logged, visualized in monitoring dashboards, or forwarded to other components of the overall security framework for further analysis and correlation.

4.6 Scalability and Security Considerations

Scalability Considerations

The proposed component must be capable of operating effectively in healthcare environments where numerous medical imaging devices generate continuous network communication. In modern hospitals, multiple imaging modalities such as CT scanners, MRI machines, ultrasound devices, and radiology workstations frequently transmit large volumes of medical images through PACS systems. Therefore, the anomaly detection component should be designed to handle increasing traffic loads without affecting system performance. Scalability is supported through the use of a modular architecture that separates packet capture, preprocessing, feature extraction, and anomaly detection stages. Efficient feature extraction techniques and lightweight machine learning models are also used to ensure that the system can process network traffic efficiently, even when the number of devices and communication sessions increases.

Security Considerations

Since this component processes healthcare network traffic data, it must ensure that sensitive information is handled securely throughout the monitoring and analysis process. Medical imaging environments contain critical patient-related information and operational data, which means that proper security mechanisms must be implemented to prevent unauthorized access or misuse of collected traffic data. The system must therefore incorporate several safeguards to protect the integrity, confidentiality, and availability of monitored data.

- secure storage of captured traffic metadata
- access control for dashboards and results
- encryption of sensitive outputs where appropriate
- anonymization of identifiers if using real-like datasets
- audit logging for access and changes

These safeguards help ensure that the monitoring component itself does not introduce additional security risks while analyzing healthcare network traffic.

5. USER REQUIREMENTS

Stakeholders

Hospital IT Administrators

Hospital IT administrators are responsible for maintaining the technical infrastructure that supports medical imaging systems, PACS servers, and related hospital networks. They need this component to monitor DICOM communication, identify abnormal traffic behavior early, and ensure that imaging services remain stable and secure.

Healthcare Security Analysts

Healthcare security analysts are responsible for detecting, investigating, and responding to suspicious activities within the healthcare network. They use the anomaly alerts generated by this component to identify possible cyber threats affecting DICOM traffic and to support faster security investigations.

Radiology System Managers

Radiology system managers oversee the day-to-day operation of radiology systems and imaging workflows within the hospital. They benefit from this component because it helps identify suspicious communication patterns that may disrupt image transfer, retrieval, or storage processes in PACS environments.

Healthcare Management / Decision Makers

Healthcare management and decision makers are responsible for ensuring service continuity, patient safety, and organizational risk reduction. Although they may not directly use the system, they benefit from improved security visibility, reduced operational disruption, and stronger protection of critical healthcare infrastructure.

Functional requirements (numbered)

The functional requirements specify what exact functionality is expected from the proposed component.

FR1: The system shall capture DICOM network traffic from PACS-related communication channels.

This requirement ensures that the system can monitor the communication exchanged between medical imaging devices, radiology workstations, and PACS servers. Capturing this traffic is the first step in observing normal and abnormal behavior within the medical imaging network.

FR2: The system shall preprocess raw packet data into structured traffic records.

Raw packet captures are difficult to analyze directly because they contain low-level network information in an unorganized form. Therefore, the system must convert these packets into structured records so that meaningful analysis can be performed efficiently.

FR3: The system shall extract relevant behavioral features from DICOM traffic.

This requirement focuses on deriving useful characteristics such as packet size, communication frequency, session duration, and request patterns from the captured traffic. These features are important because they help represent the behavior of DICOM communications in a format suitable for machine learning analysis.

FR4: The system shall apply a machine learning-based anomaly detection model to the extracted features.

The extracted features will be passed into a trained machine learning model to identify whether the traffic behavior is normal or suspicious. This allows the system to move beyond static rule-based detection and identify unusual patterns that may not match known attack signatures.

FR5: The system shall detect abnormal communication patterns in DICOM operations.

This requirement enables the system to identify suspicious usage of DICOM operations such as C-STORE, C-FIND, and C-MOVE. Detecting such abnormal patterns is essential because unusual imaging communication may indicate unauthorized access, misuse, or possible cyberattacks.

FR6: The system shall generate alerts when suspicious traffic behavior is identified.

Once an anomaly is detected, the system must produce a clear alert to inform administrators or security analysts. These alerts help ensure that potentially harmful activities are noticed early and can be investigated before causing serious damage.

FR7: The system shall store anomaly results for analysis and reporting.

This requirement ensures that detected anomalies and related information are saved for future review, auditing, and performance evaluation. Storing results also supports incident investigation and helps in generating security reports for healthcare administrators.

FR8: The system shall support integration of anomaly outputs with the broader hybrid framework.

The anomaly detection results from this component should be usable by other components in the overall hybrid healthcare security framework. This integration is important because it allows network-level findings to contribute to a larger cross-layer view of threats in the healthcare environment.

Non-functional requirements

Non-functional requirements specify how well the system should be and any constraints involved in its use.

1. Security Requirements

NFR1: The system shall protect captured network data against unauthorized access.
NFR2: The system shall restrict access to alert outputs and dashboards through authentication mechanisms.
NFR3: Sensitive monitoring data shall be handled securely during storage and transfer.

2. Performance Requirements

NFR4: The system shall process captured traffic efficiently enough for near real-time anomaly detection.
NFR5: The system shall handle a reasonable volume of imaging traffic without excessive delay.

3. Usability Requirements

NFR6: The alert outputs shall be understandable to hospital IT and security staff.
NFR7: The reporting interface shall clearly display suspicious flows and anomaly summaries.

4. Scalability Requirements

NFR8: The system shall support additional imaging devices and increasing network traffic volumes with minimal architectural changes.

5.Reliability Requirements

NFR9: The component shall remain stable during continuous monitoring sessions and repeated testing.

6. COMMERCIALIZATION PLAN

The proposed component has strong commercialization potential in healthcare cybersecurity, especially in environments where medical imaging infrastructure must be monitored continuously.

1 Target market
Hospitals, radiology centers, diagnostic laboratories, private healthcare providers, and managed security service providers supporting healthcare clients.

2 Value proposition
The component provides a specialized anomaly detection capability for DICOM and PACS environments, enabling early discovery of suspicious imaging traffic that general network security tools may miss.

3 Revenue model
The solution could be offered as an annual subscription, per-site licensing model, or as part of a broader managed healthcare security platform.

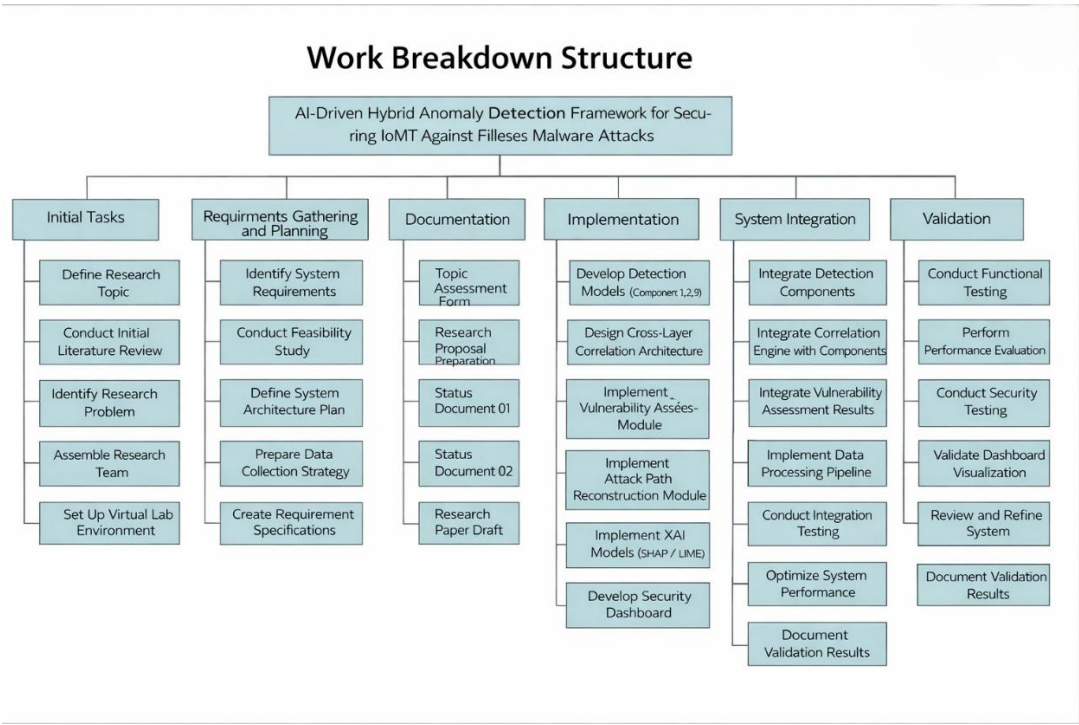
4 Cost estimation
Deployment costs would mainly involve a monitoring server, storage, integration effort, and ongoing maintenance.

5 Pricing strategy
Pricing can be introduced in tiers such as basic DICOM monitoring, advanced anomaly analytics, and enterprise multi-site deployment.

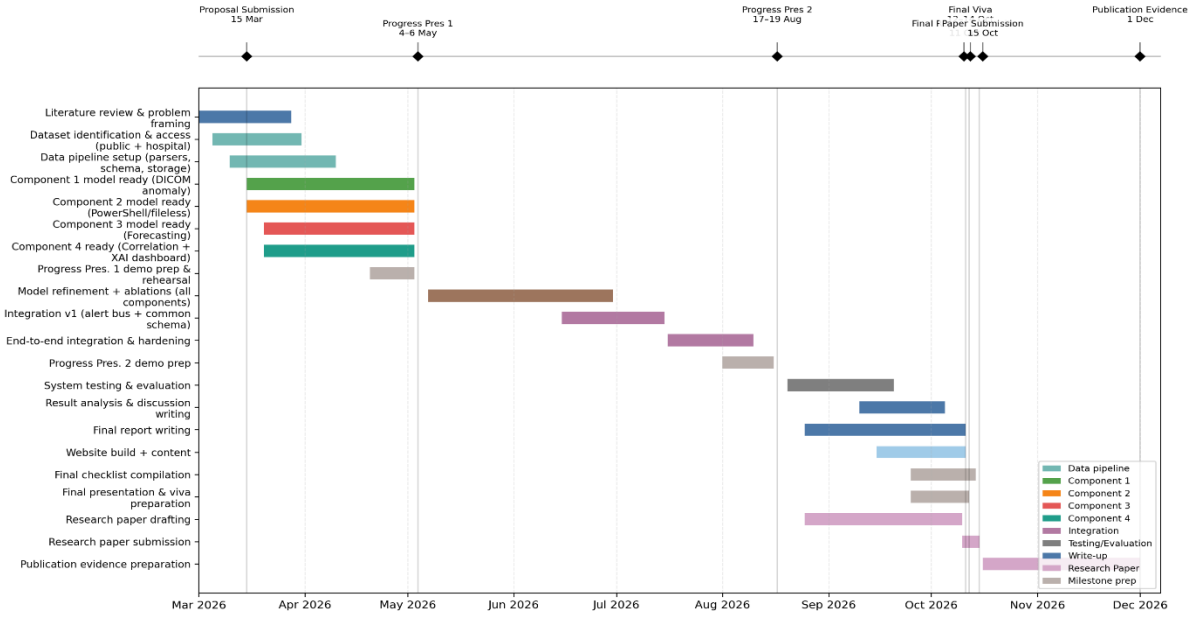
6 Competitive advantage
Its competitive advantage lies in being healthcare-specific, lightweight, and protocol-aware, unlike generic network anomaly tools.

7 IP considerations
Any original feature engineering pipeline, anomaly scoring strategy, and healthcare-specific visualization logic could be considered for protection, while ensuring compliance with open-source license usage.

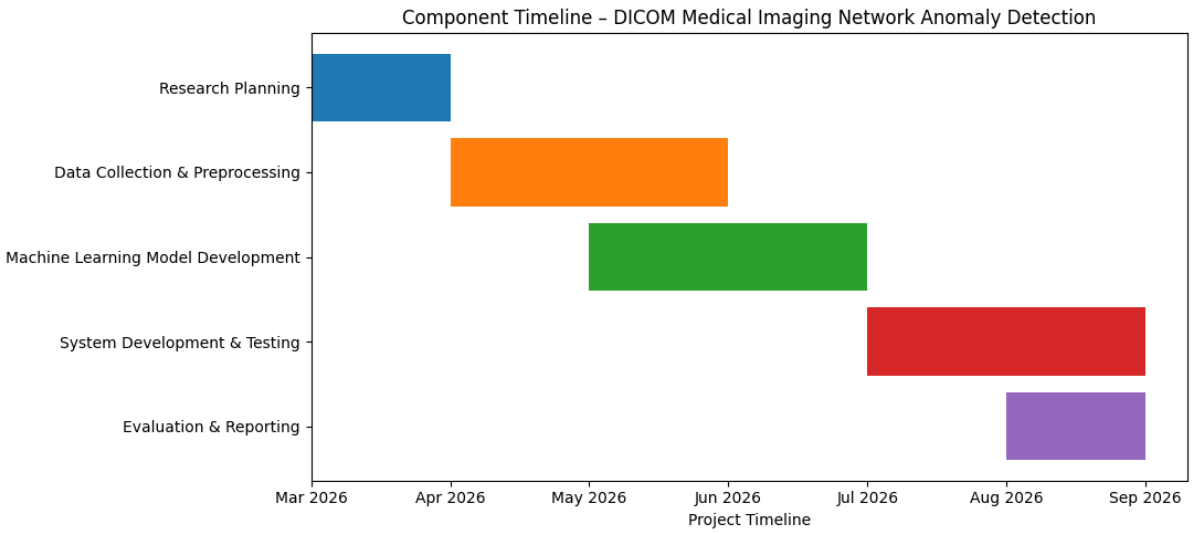
7. WORK BREAKDOWN STRUCTURE (WBS)



9. GANTT CHART (Full Project)



My Component



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